

SEMESTER END EXAMINATIONS – AUGUST 2024

Program	: B.E. – Information Science and Engineering	Semester	: IV
Course Name	: Database Management Systems	Max. Marks	: 100
Course Code	: IS44 / IS45(00)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define the following terms:- CO1 (05)
i) Database ii) Meta Data iii) Schema iv) Keys v) user view.
 - Explain the difference between external, internal and conceptual schemas. How are these related to the concepts of Data independence. CO1 (07)
 - The library provides books to borrowers. Each book is described by title, edition, and year of publication, and is uniquely identified using the ISBN. Each borrower is described by his or her name and address and is uniquely identified using a borrower number. The library provides one or more copies of each book and each copy is uniquely identified using a copy number, status indicating if the book is available for loan, and the allowable loan period for a given copy. A borrower may loan one or many books, and the date each book is loaned out and is returned is recorded. Loan number uniquely identifies each book loan. Draw an ER diagram for the above requirements. CO1 (08)
- List the different types of database end users? Discuss the main activities of each. CO1 (06)
 - A large organization has several parking lots, which are used by staff. CO1 (08)
 - Each parking lot has a unique name, location, capacity, and number of floors (where appropriate).
 - Each parking lot has parking spaces, which are uniquely identified using a space number.
 - Members of staff can request the sole use of a single parking space. Each member of staff has a unique number, name, telephone extension number, and vehicle license number.
 - Represent all the requirements described in parts (i), (ii), (iii), and (iv) as a single ER model. Provide any assumptions necessary to support your model.
 - Discuss the main characteristics of the database approach and how it differs from traditional file systems. CO1 (06)

UNIT - II

- Consider the following requirements for a simple database for the Pro Kabaddi League (PKL): CO2 (07)
 - The PKL has many teams,
 - Each team has a name, a city, a coach, a captain, and a set of players,
 - Each player belongs to only one team,
 - Each player has a name, a position (such as right and left corners, right and left covers, raiders, defenders etc.), and a set of injury records,

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- A team captain is also a player,
- A game is played between two teams (referred to as host_team and guest_team) and has a date (such as May 11th, 2023) and a score (such as 4 or 2).

Construct a clean and concise ER diagram for the PKL database. List your assumptions and clearly indicate the cardinality mappings and any role indicators in your ER diagram. Specify all the key attributes clearly.

- b) Show UNION, INTERSECTION, and MINUS operations in relational algebra with examples. CO2 (06)
- c) Outline the key concepts of the Relational Model, emphasizing relational model constraints and schemas. CO2 (07)
4. a) Explore the unary relational operations: SELECT and PROJECT with examples. CO2 (04)
- b) Consider the following schema: CO2 (08)
- Suppliers (sid: integer, name: string, address: string)
- Parts (pid: integer, pname: string, color: string)
- Catalog (sid: integer, pid: integer, cost: real)
- Write the following queries in relational algebra.
- Find the *names* of suppliers who supply some red part.
 - Find the *sids* of suppliers who supply some red or green part.
 - Find the *sids* of suppliers who supply some red part or are at 221 Packet Street.
 - Find the *sids* of suppliers who supply every red part.
- c) Briefly explain different constraints of relational databases with proper examples. CO2 (08)

UNIT - III

5. a) Consider the following table with the details of products sold in a shop for two days. CO3 (08)

PRODUCTS

Date	ItemName	Unit_Sold
9/3/24	BOOKS	200
9/03/24	PENDRIVE	100
10/3/24	BOOKS	100
10/3/24	PENDRIVE	200
10/3/24	SHARPENER	75

Construct SQL queries for the following scenarios:

- Display the total number of items sold for each item name
 - Display the item names and total number of items sold for only those items where total of sold items is greater than 1000
 - Display the table sorted in the descending order on the number of quantities sold.
 - Display on which dates BOOKS are sold.
- b) Explain the different aggregate functions supported by SQL. CO3 (06)
- c) Explain the use of IN operator in SQL Queries with one example. CO3 (06)

6. a) Explain with one example any five clauses used in SQL. CO3 (10)
b) Consider the following Relation Schema, CO3 (10)
EMPLOYEE (empno,name,office,age)
BOOKS (isbn,title,authors,publisher)
LOAN (empno,isbn,date)
Write the following queries in SQL.
i) Find the names of employees who have borrowed a book published by McGraw-Hill.
ii) Find the names of employees who have borrowed all books published by McGraw-Hill.
iii) Display the number of books borrowed by an employee
iv) List all the books which are borrowed on specific date
v) Display the average age of employee who have borrowed same books.

UNIT- IV

7. a) Explain any three Informal design guidelines for relational database design. CO4 (06)
b) Define Functional dependency. With one example for each, explain the types of Functional dependencies. CO4 (06)
c) Suppose that you have a relation R (A, B, C, D, E, F) with FDs $AB \rightarrow CD$ and $D \rightarrow EF$. Decompose R into a set of relations, each of which is in BCNF, or state that R is already in BCNF. CO4 (08)
8. a) Why is Normalization important? When you say that the relation schema satisfies First Normal form. Give an example. CO4 (08)
b) Given a relation R (A, B, C, D) and Functional Dependency set $FD = \{AB \rightarrow CD, B \rightarrow C\}$, determine whether the given R is in 2NF? If not, convert it into 2 NF. CO4 (06)
c) What are Anomalies in databases. Illustrate the scenario of the deletion anomaly. CO4 (06)

UNIT-V

9. a) Discuss the need for a checkpoint entry in the log. List the actions of taking a checkpoint. CO5 (10)
b) Compare binary locks to exclusive /shared locks. Why is the latter type of locks preferable? CO5 (10)
10. a) Draw a state diagram, and discuss the typical states that a transition goes through during execution. CO5 (10)
b) What is ARIES? Discuss its three phases. On what principles is it based on? CO5 (10)
